

Year 11 Mathematics (2u) – Half Yearly Examination 2006

Question 1	Marks
a. Solve $ 2x + 6 = 3x - 1$.	3
b. Find the domain and range for $y = \frac{1}{\sqrt{9 - x^2}}$.	2
c. Find the value of n for $\frac{9^n \cdot 2^{n-1}}{36^n} = 1$.	3
d. Find the derivative of $\frac{4x + \sqrt{x}}{x}$.	2
e. Evaluate $\lim_{x \rightarrow -1} \frac{x + 1}{3x^2 + 5x + 2}$.	2
f. Simplify and express with a rational denominator in its simplest form : $\frac{\sqrt{5}(3 - \sqrt{20})}{\sqrt{5} - 2(\sqrt{5} - 10)}$	3

Question 2 (Start a new page)

- | | |
|--|----------|
| a. Is the line $x + y = 3$ tangent to the circle $2x^2 + 2y^2 = 9$?
Give reasons. | 2 |
| b. i. Find, by first principles, the derivatives of $f(x) = x^2$. | 2 |
| ii. Find the equation of tangent to $f(x) = x^2$ at $x = 3$. | 2 |
| c. Let A and B be the fixed points $(-1,0)$ and $(2,0)$ and let P be the variable point (x, y) , | |
| i. Suppose that P moves so that $PA = 2PB$. Deduce that P moves on a circle. | 2 |
| ii. Find the centre and radius of the circle. | 2 |
| d. i. If $\tan \theta = c$, for $90^\circ < \theta < 180^\circ$, find the value of $\sin \theta$ in terms of c . | 2 |
| ii. Solve for x : $\cos^2 x = \frac{3}{4}$, $0^\circ \leq x \leq 360^\circ$ | 3 |

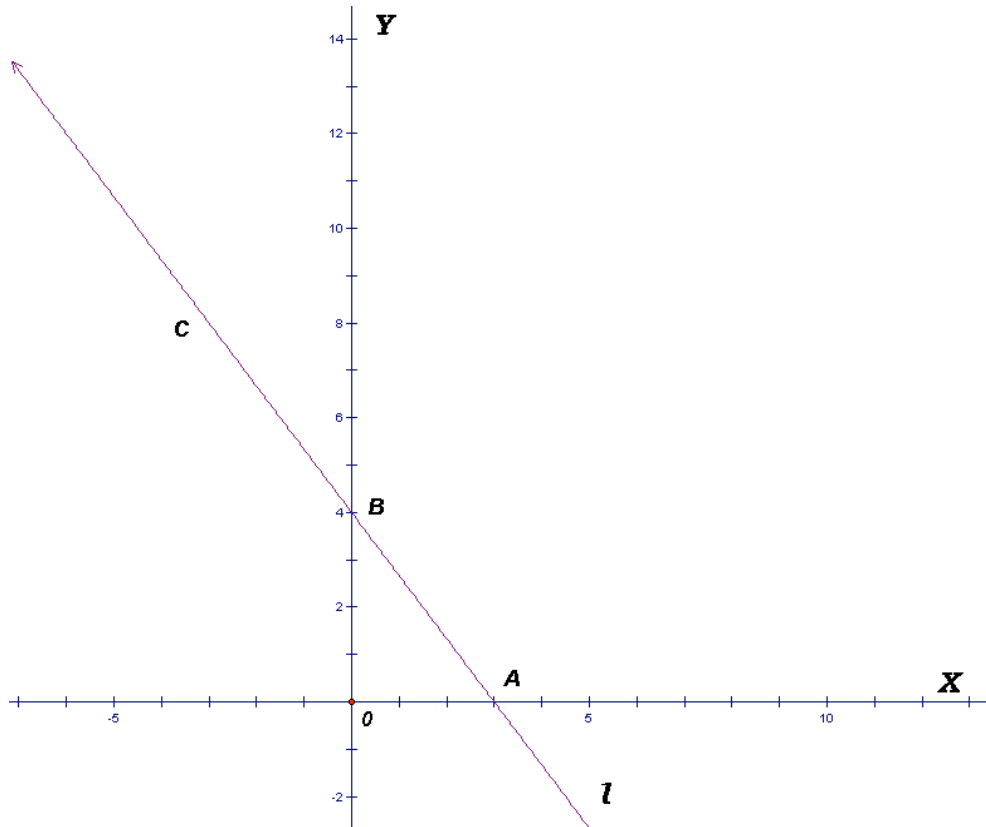
Question 3 (Start a new page)

- | | |
|---|----------|
| a. From a point A on a straight road running north and south, the bearing of a point C is N 38° E. From B , 9570 metres north of A , the bearing of C is S 41° E. Calculate | |
| i. the distance from A to C (to the nearest metre). | 3 |
| ii. the shortest distance from C to the road (to the nearest metre). | 2 |

Question 3 (continued)

b.

Marks



In the diagram above, the line l cuts the x axis at $A(3,0)$ and y axis at $B(0,4)$.

C is a point on l such that $AB=BC$.

$D(3,8)$ is on the line k where the line k (not shown on diagram) has equation :
 $4x + 3y - 36 = 0$.

Copy or trace the diagram into your writing booklet.

- i. Find the equation of l . 2
- ii. Draw the graph of k in your diagram indicating where it cuts the axes. 1
- iii. Show that the coordinates of C is $(-3, 8)$. 1
- iv. Find the perpendicular distance of the point C from the line k : $4x + 3y - 36 = 0$. 2
- v. M is the midpoint of CD . Show the equation of a circle centre D and radius DM has the equation $x^2 - 6x + y^2 - 16y + 64 = 0$. 2
- vi. In your diagram shade the region that satisfies both $x^2 - 6x + y^2 - 16y + 64 \leq 0$ and $4x + 3y - 36 \geq 0$ simultaneously. 2

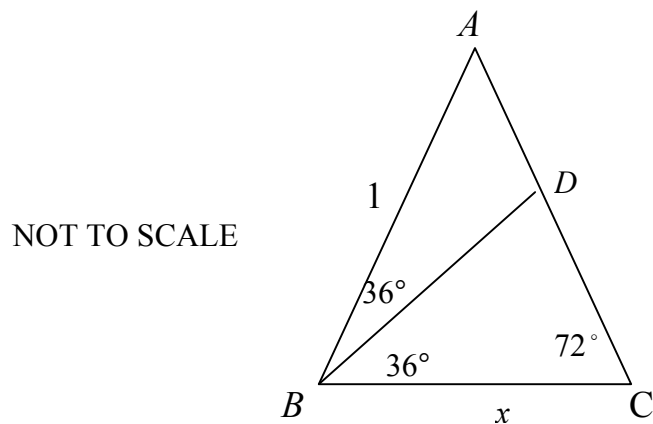
Question 4 (Start a new page)

Marks

- a. Graph $y = \frac{x}{x+1}$, showing all the essential features.

3

b.



In the diagram above, ABC is an isosceles triangle where $\angle ABC = \angle BCA = 72^\circ$ and $AB = AC = 1$. Angle ABC is bisected by BD and $BC = x$.

Copy the diagram into your answer booklet.

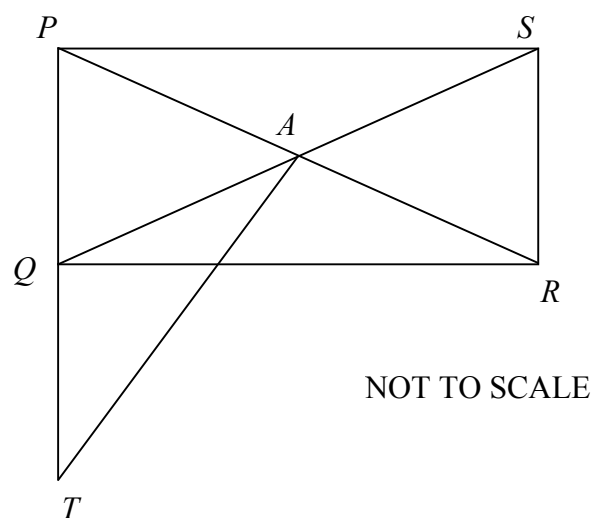
- i. Show that triangles ABC and BCD are similar.

2

- ii. By using part i, find the exact value of x .

4

c.



In the diagram above, $PQRS$ is a rectangle. Diagonals PR and SQ meet at A . Q is the midpoint of PT and $PT = PR$.

Copy the diagram into your answer booklet.

- i. Prove $\triangle PRQ \cong \triangle TRQ$.

2

- ii. Hence, prove $AR \perp AT$.

4

END of PAPER

Solutions to 2006 2u T2

(1a) $2x+6=3x-1$ or $2x+6=1-3x$

$x=7$ 1m or $x=-1$ 1m

check $|2x+6|=3x-1$

If $x=7$ LHS $= |2(7)+6|=20$

RHS $= 3(7)-1=20$

$\therefore x=7$

If $x=-1$

LHS $= |2(-1)+6|=4$

RHS $= 3(-1)-1=-4$

$\therefore x \neq -1$

$\therefore x=7$ only 1m

b) $D: -3 < x < 3$ 1m

$R: \frac{1}{3} \leq y$ 1m

c) $\frac{3^{2n} \cdot 2^{n-1}}{2^{2n} \cdot 3^{2n}} = 1$ 1m

$2^{n-1} = 2^{2n}$ 1m

$\therefore -1 = n$ 1m

d) $\frac{4x + \sqrt{x}}{x} = 4 + \frac{1}{\sqrt{x}}$ 1m

$\therefore \frac{d}{dx} \left(4 + \frac{1}{\sqrt{x}} \right) = -\frac{1}{2\sqrt{x}}$ 1m

e) $\lim_{x \rightarrow -1} \frac{x+1}{(3x+2)(x+1)} = \lim_{x \rightarrow -1} \frac{1}{3(-1)+2} = -1$ 1m

f) $\frac{3\sqrt{5} - \sqrt{100}}{\sqrt{5} - 2\sqrt{5} + 20} = \frac{3\sqrt{5} - 10}{20 - \sqrt{5}} \times \frac{20 + \sqrt{5}}{20 + \sqrt{5}} = \frac{60\sqrt{5} - 200 + 15 - 10\sqrt{5}}{400 - 5}$ 1m
 $= \frac{50\sqrt{5} - 185}{395} = \frac{10\sqrt{5} - 37}{79}$ 1m

Q2a) $y = 3 - x$

$2x^2 + 2(3-x)^2 = 9$

$2x^2 + 2(9 - 6x + x^2) = 9$

$4x^2 - 12x + 18 - 9 = 0$

$4x^2 - 12x + 9 = 0$

$(2x-3)^2 = 0$ 1m

$x = \frac{3}{2}$ (double root)

yes, $x+y=3$ is tangent to $2x^2+2y^2=9$ 1.
 because it touches the circle at only 1 point.

b) c) $f(x) = x^2$

$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$ 1m
 $= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = 2x$ 1m

(ii) Eq of tangent to $f(x) = x^2$ at $x=3$, $y=9$

$y-9 = 2(3)(x-3)$ 1m

$y-9 = 6x-18$

$y = 6x - 9$ 1m

c) $(PA)^2 = (x+1)^2 + y^2$ $P(B)^2 = (x-2)^2 + y^2$

$PA = 2PB \therefore (PA)^2 = 4(PB)^2$

$(x+1)^2 + y^2 = 4[(x-2)^2 + y^2]$ 1m

$x^2 + 2x + 1 + y^2 = 4(x^2 - 4x + 4 + y^2)$

$x^2 + 2x + 1 + y^2 = 4x^2 - 16x + 16 + 4y^2$

$-15 = 3x^2 - 18x + 3y^2$ 1m

$$\text{cii)} \quad -15+27 = 3(x^2-6x+9) + 3y^2$$

$$12 = 3(x^2-6x+9) + 3y^2$$

$$4 = (x-3)^2 + y^2$$

$\therefore$ centre of circle is $(3, 0)$, radius = 2 unit 1m

di) $\tan \theta$ and $\sin \theta$ have different signs for $90^\circ < \theta < 180^\circ$

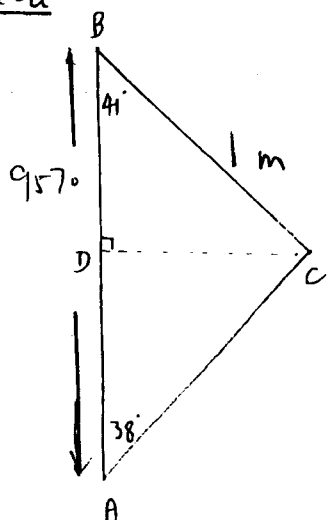
$$\therefore \sin \theta = \frac{-c}{\sqrt{1+c^2}} \quad \#$$

1m for "sign"

1m for correct answer

$$\text{ii)} \quad \cos x = \pm \frac{\sqrt{3}}{2} \quad 1m \quad \therefore x = \underline{30^\circ, 150^\circ, 210^\circ, 330^\circ}$$

23a



$$\angle BCA = 180^\circ - 41^\circ - 38^\circ = 101^\circ$$

$$\frac{9570}{\sin 101^\circ} = \frac{AC}{\sin 41^\circ} \quad 1m$$

$$AC = \frac{9570 \sin 41^\circ}{\sin 101^\circ} = 6396 \text{ (nearest metre)} \quad 1m$$

$$CD \perp AB$$

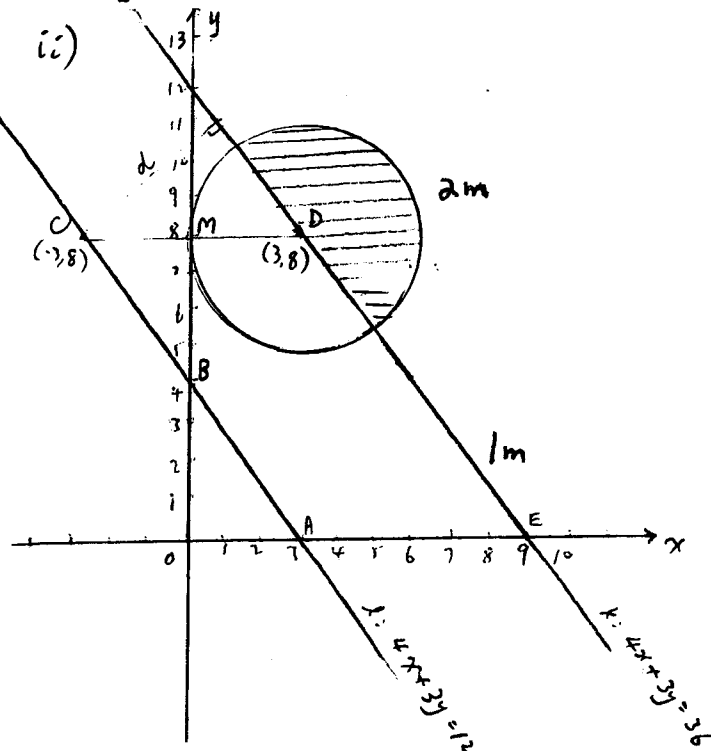
$$CD = AC \sin 38^\circ = 3938 \text{ (nearest metre)} \quad 1m$$

Q 36 Eq of line l:

$$\text{i)} \quad y - 0 = (x - 3) \left[\frac{4}{-3} \right] \quad 1m$$

$$\begin{aligned} -3y &= 4x - 12 \\ 12 &= 4x + 3y \quad 1m \end{aligned}$$

ii)



$$\text{iii)} \quad \text{Let } C = (x, y) \quad \left. \begin{aligned} \frac{x+3}{2} &= 0 \quad \therefore x = -3 \\ \frac{y+0}{2} &= 4 \quad \therefore y = 8 \end{aligned} \right\} 1m$$

$$\therefore C = (-3, 8)$$

$$\text{iv)} \quad d = \frac{|4(-3) + 3(8) - 36|}{\sqrt{4^2 + 3^2}} = \frac{|-12 + 24 - 36|}{5} = \frac{24}{5} \quad 1m$$

$$\text{v)} \quad D \text{ centre of circle} = (3, 8) \quad DM = 3 \text{ (radius)}$$

$$(x-3)^2 + (y-8)^2 = 3^2 \quad 1m$$

$$x^2 - 6x + 9 + y^2 - 16y + 64 = 9$$

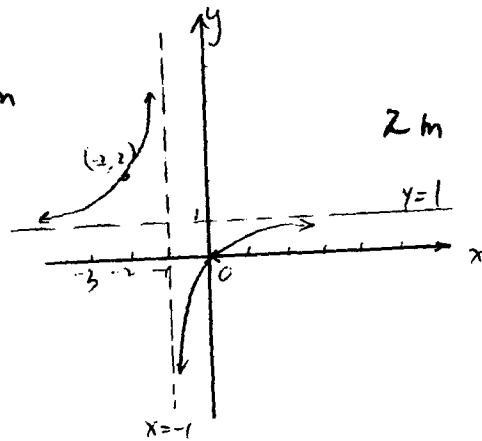
$$x^2 - 6x + y^2 - 16y + 64 = 0 \quad 1m$$

1.

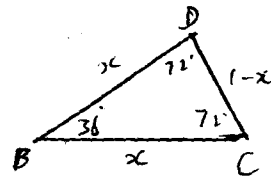
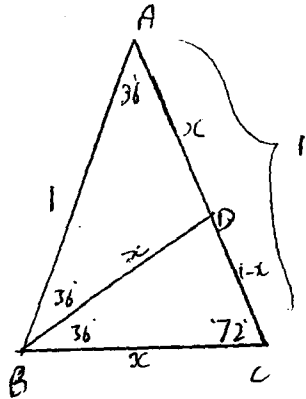
$$\text{Asy: } \begin{cases} x = -1 \\ y = 1 \end{cases}$$

Intercept $x=0, y=0$

* any other pt on the other branch $-\frac{1}{2}$



b)



m i) In $\triangle ABC$, $\angle ACB = \angle ABC = 72^\circ$ (given)

m { In $\triangle BCD$, $\angle DCB = 72^\circ$ (given)
 $\angle DBC = 36^\circ$ (given)

m $\therefore \angle BDC = 180^\circ - 36^\circ - 72^\circ = 72^\circ$ (angle sum of triangle)

$\therefore \angle DCB = \angle BDC = 72^\circ$

m $\therefore \triangle ABC \cong \triangle BCD$ (equiangular)

ii) $\angle BAC = 36^\circ$ (given)

$\angle CAB = 180^\circ - 72^\circ - 72^\circ = 36^\circ$ (angle sum of triangle ABC)

$AD = BD$ (sides opposite equal angles are equal)

Similarly $BC = BD$

1 m $\frac{AB}{BC} = \frac{BC}{DC}$ (corresponding sides of similar triangles ABC, BCD)

Let $AD = BD = BC = x$, $DC = 1 - x$

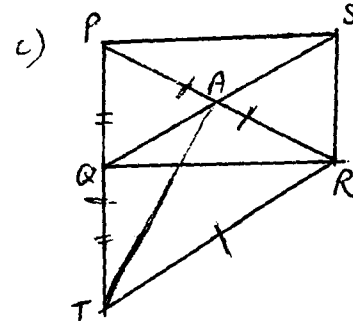
$$1 \text{ m } \frac{1}{x} = \frac{x}{1-x}$$

$$1-x = x^2$$

$$0 = x^2 + x - 1$$

$$\frac{1}{2} \text{ m } x = \frac{-1 \pm \sqrt{1-4(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

$$1 \text{ m } \text{but } x > 0 \therefore x = \frac{\sqrt{5}-1}{2} \#$$



To prove: i) $\triangle PRQ \cong \triangle TRQ$

Proof In $\triangle PRQ, \triangle TRQ$,

$\frac{1}{2}$ $PQ = TQ$ (Q is mid pt of PT)

$\frac{1}{2}$ RQ is common

$\frac{1}{2}$ $\angle PQR = 90^\circ$ (angle of rectangle)

$\frac{1}{2}$ $\therefore \angle TQR = 180^\circ - 90^\circ = 90^\circ$ (angle sum of straight line PQT)

$\frac{1}{2}$ $\therefore \triangle PRQ \cong \triangle TRQ$ (SAS)

ii) To prove $AR \perp AT$:

$\frac{1}{2}$ $PR = TR$ (corresponding sides of congruent triangles PRQ, TRQ)

In $\triangle PAT$ and $\triangle RAT$

$\frac{1}{2}$ $PT = TR$ (given)

$\frac{1}{2}$ AT is common

$\frac{1}{2}$ $PA = RA$ (diagonal of rectangle PSRQ bisect each other)

$\frac{1}{2}$ $\triangle PAT \cong \triangle RAT$ (SSS)

$\frac{1}{2}$ $\angle PAT = \angle RAT$ (corresponding angles of congruent triangles PAT, RAT)

$\frac{1}{2}$ $\angle PAT + \angle RAT = 180^\circ$ (angle sum of straight line PAR)

$\frac{1}{2}$ $\therefore \angle PAT = \angle RAT = 180^\circ \div 2 = 90^\circ$

$\therefore AR \perp AT$